

Health & Safety

Standard Special works and deep foundations



Reference

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BYTP	REF	H&S	2228	A	EN

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In the event of technical impossibility of creating superficial foundations (presence of compressible soils), deep foundations allow to transfer the structural loads of the structures into the better quality layers at depth.

The standards for special works and deep foundations presented in this document aim to provide a practical guide to define the means to be planned and deployed.

The elements of this document are supplemented by the elements presented in the following standards:

- Standard Lifting [BYTP-REF-H&S-2210]
- Standard Fall from height prevention [BYTP-REF-H&S-2211]
- Standard Vehicles pedestrian interaction [BYTP-REF-H&S-2212]
- Standard Stability [BYTP-REF-H&S-2213]

The elements of this document also complement the elements presented in the following MOSE+:

- 5702 – Recepieous coppering
- 5834 – Recovering of cased piles using a hydraulic splitter
- 5836 – Recovering of cased piles by overflow
- 5838 – Install filling monitoring tubes for deep foundations

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1. Bored/cased/secant piles

Drilling carried out in the ground by mechanical means (auger, buckets, etc.), under the protection of a casing:

- The casing can be driven to the final depth by vibration or sunk with swaying as drilling progresses
- The borehole is partially or completely filled with concrete and reinforcement, then the casing is extracted

In the case of sufficiently coherent soil located above the water tables, simple bored piles are used for which the principle of the operation remains the same but the casing is not required.

The main risks associated with pile drilling operations are:

- Falling into boreholes
- Equipment overturning
- Falling objects
- Hit by machinery
- etc.

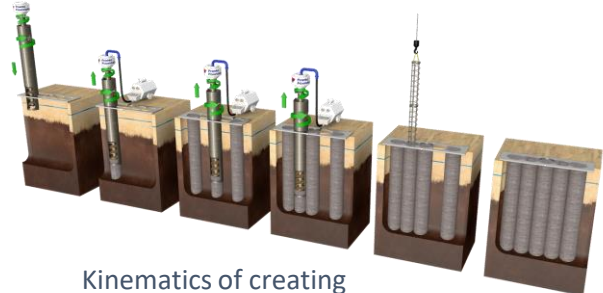
Collective protections to be implemented



Secant piles (pile wall)

The secant piles create retaining and waterproofing walls and are subject to the same specifications as mentioned previously.

- Creation of guide walls prior to drilling operations
- Coping beam



Kinematics of creating secant piles

2. Micropiles

- Drilled piles with diameter <250mm
- Drillings filled with cement grout and equipped with reinforcement
- Concreting, then the casing is extracted

The main accidents occurring during drilling operations are hand and foot injuries:

- Entrapment by rotating drill rods
- Risks associated with manual handling of drill pipes
- Risks linked to the mobilization of drills (raising/lowering the mast)
- Overturning of the drill due to ground instability

The plan of nearby networks must be up to date and the networks identified (marking-staking) before any drilling!

Collective protection to be implemented

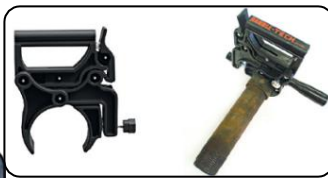
Examples of different anti-trapping systems



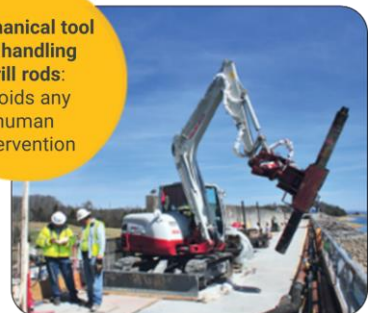
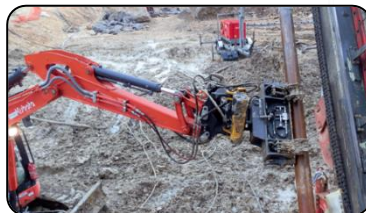
- Protective cage for rotating parts is MANDATORY for drilling equipment.
- In restricted, constrained spaces (against a wall, etc.) the protection may be removed if the risk analysis demonstrates the technical impossibility of maintaining it, and by the implementation of effective compensatory measures to avoid any risk of entanglement. : detection system, physical marking for example



Manual handling device: helps prevent fingers/hands from getting caught



Mechanical tool for handling drill rods: avoids any human intervention



3. Trimming work

Whatever the types of piles to be trimmed, it is imperative to install foam sheaths on the pending steels and sonic tubes. Also install the lifting rings for the cut parts.



Foam sheaths



lifting ring

Trimming by “Recépieux” system

Trimming using Recépieux flasks (dangerous chemical product):

- Setting up the collection tubes and flasks
- Using the expansive product
- Wearing specific PPE required (3 days minimum after concreting of the pile)
- Withdrawal possible between 2 and 72 hours after application of the expansive product



Trimming with hydraulic splitter

Trimming using a hydraulic splitter:

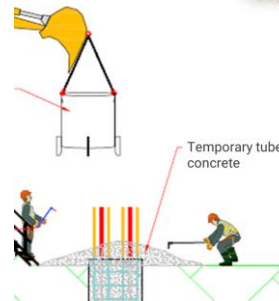
- Referenced perforator
- Bursting cylinder
- Hydraulic central
- High and low pressure hoses



Trimming by overflow

Trimming using a temporarily welded tube on the pile.

- Concreting piles
- Removal of the temporary tube
- Spreading concrete using it as cleaning concrete



Trimming with hydraulic pile breaker

The pile breaker is a device specially designed for cutting circular piles. It can also be adapted to piles with rectangular sections.

It consists of a crown equipped with knives and hydraulic cylinders, the axes of which converge towards the axis of the pile.



4. Injections / field treatments

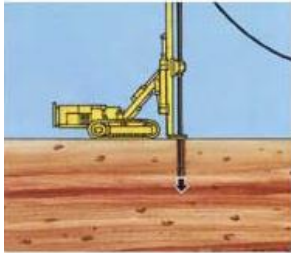
Ground injections essentially make it possible to improve soil characteristics

- Treatment injections
- Jet grouting
- Soil mixing
- etc.

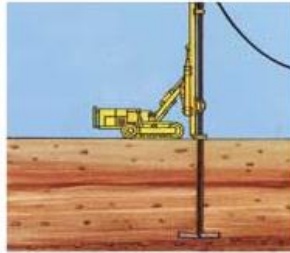
The plan of nearby networks must be up to date, the networks identified (marking-staking) before any drilling!

How Jet-grouting Works

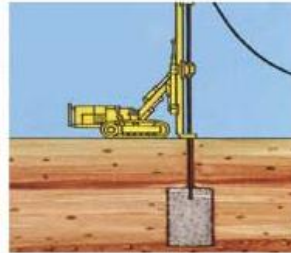
Step 1: Start drilling



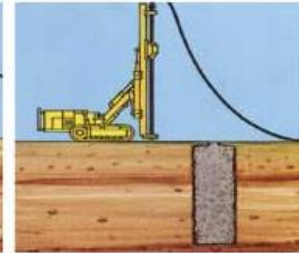
Step 2: Start jet-grouting from the bottom



Step 3: Pull up the rotating jet



Result: A round column consisting of water/cement mix and in-situ soil



Use of drills / general installations

- Drills must comply with current regulations
- Protective cage for rotating parts is MANDATORY on drilling equipment
- In restricted, constrained spaces (against a wall, etc.) the protection may be removed if the risk analysis demonstrates the technical impossibility of maintaining it, and by the implementation of effective compensatory measures to avoid any risk of entanglement : detection system, physical marking for example



Tube storage and spoils management

- Store the tubes as close as possible to the workstations
- Mechanize the gripping of the tubes for installation on the drill when technically possible (limit manual handling)
- The tube preparation workshops are designed so that the workers work at eye level
- Ensure the availability of a water point/tank storage near the workstation
- Provide a buffer stock of spoils linked to drilling (allows you to keep the work area clean)



5. Diaphragm walls

A diaphragm wall (D wall) is a reinforced concrete wall made in the ground and composed of a juxtaposition of elements interdependent on each other, called panels.

The technique used consists of drilling each element, equipping it with reinforcement cages and concreting it using the ground as formwork (with end sheet piles).



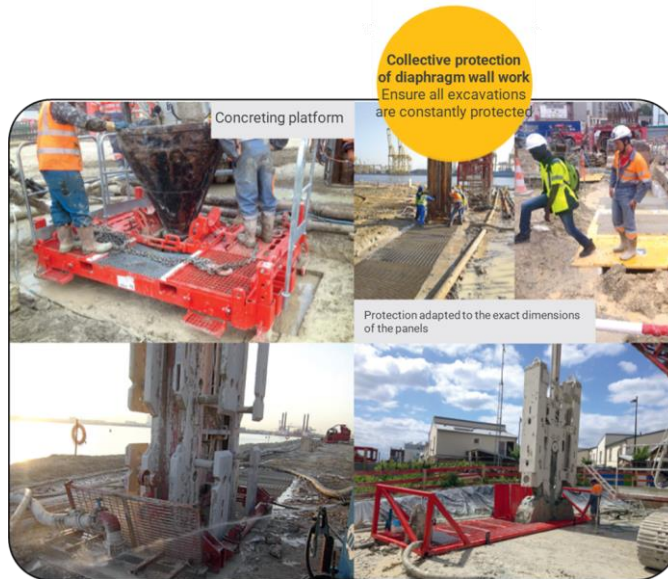
Bentonite

- As the land is extracted, it is “replaced” by a mixture of water and bentonite (a treated clay)
- Bentonite being denser than water, it counteracts the hydrostatic pressure and prevents the excavated walls from collapsing.
- Provide adequate collective protection on the guide walls on the interior side



Installation of diaphragm walls
(bentonite silos, sludge treatment, etc.)

Collective protection to be implemented



Collective protections (ring) of the panel during excavation placed on the guide walls

Stability

Stability of the diaphragm wall structure ensured by the buttons of the infrastructure.

